

• Realistic · Reproducible · Deep Research Evaluation

DR³-Eval

Towards Realistic and Reproducible Deep Research Evaluation

 View Leaderboard

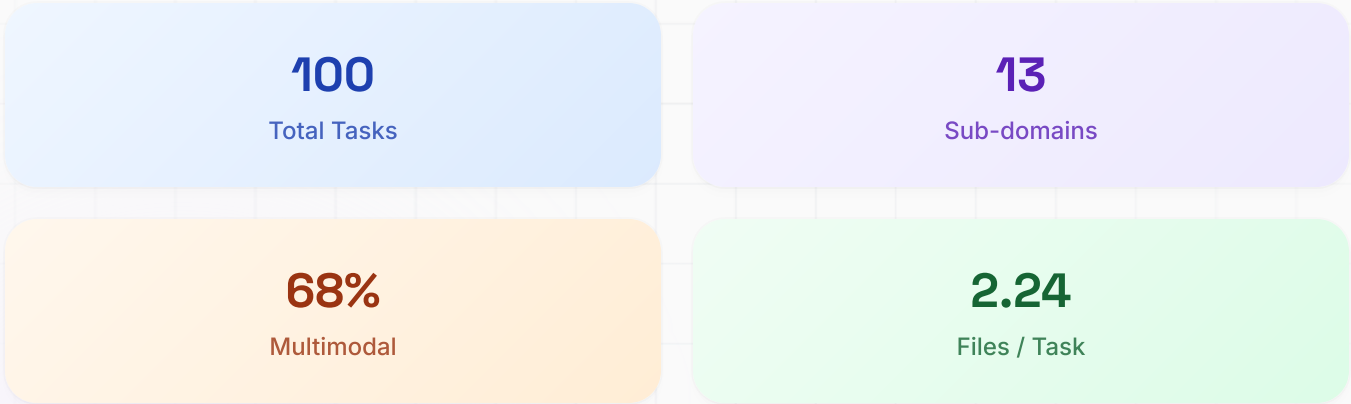
 Arxiv Paper

 GitHub

 Dataset

 Submit

100 Tasks 13 Sub-domains 68% Multimodal 2.24 Files/Task



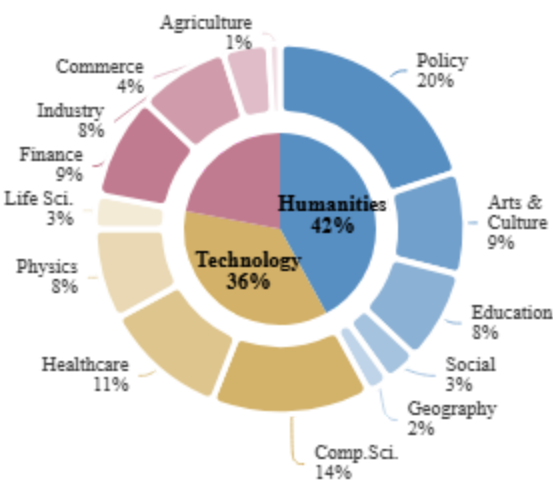
Dataset Statistics

100 tasks (50 EN + 50 ZH), 13 sub-domains, 68% multimodal input.

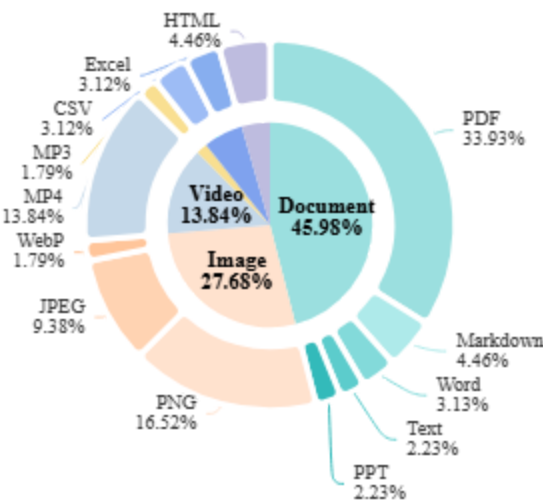
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Domain Distribution



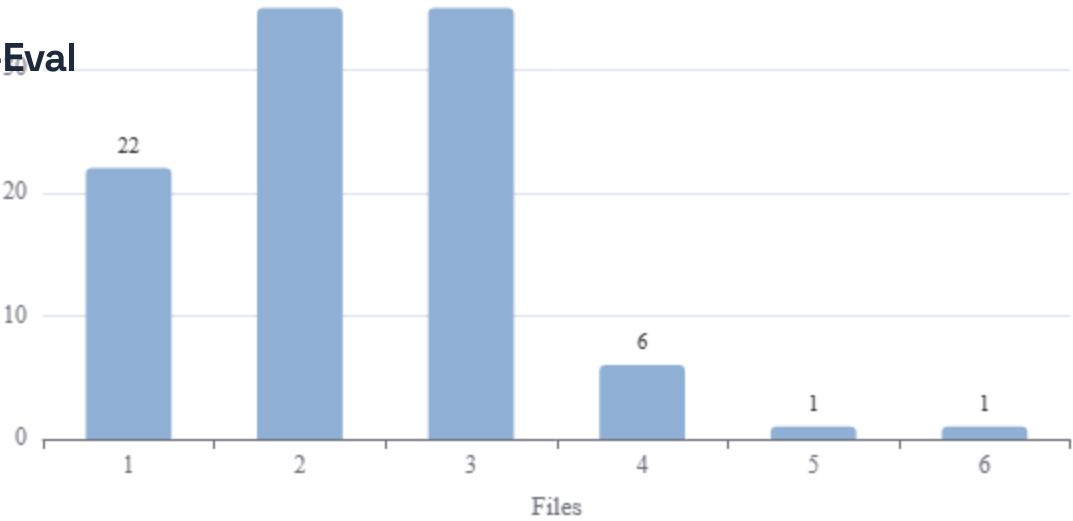
- File Type Distribution



- User Files Per Task



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Evaluation Metrics

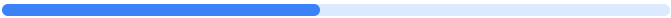
5 complementary metrics across two dimensions to assess deep research agent performance.

• 🔍 Information Seeking



IR — Information Recall

47.6

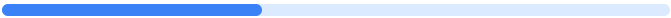


Coverage of key insights from user files and sandbox corpus.

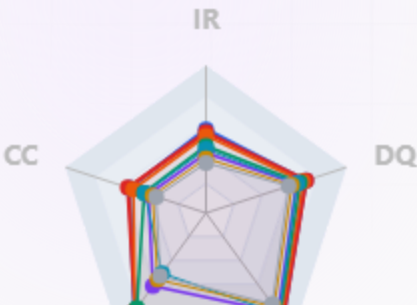


CC — Citation Coverage

39.0



Overlap between cited and required source documents.



• 📄 Report Generation

D3 DR³-Eval

- Claude Sonnet 4
- GLM-4.7
- GLM-4.6
- Gemini-2.5-Pro
- GPT-4.1
- Qwen3-235B
- Qwen3-32B
- Qwen2.5-72B



FA — Factual Accuracy

67.3

Factual correctness of claim-source pairs via entailment.



IF — Instruction Following

83.4

Whether report satisfies each requirement in a verified checklist.



DQ — Depth Quality

66.5

Expert judge evaluates analytical substance on 1-10 scale.



Leaderboard

Evaluation results of 8 state-of-the-art LLMs across 64k/128k/512k sandbox corpus scales.

All Scales

64k

128k

512k

Model	IR _{UF}			IR _{SC}			CC			FA			IF		
	64k	128k	512k	64k	128k	512k	64k	128k	512k	64k	128k	512k	64k	128k	512k
Claude Sonnet 4	58.8	60.4	60.8	55.3	46.6	41.8	64.7	54.8	48.5	87.0	82.7	82.1	87.4	89.2	88.1
GLM-4.7	55.7	55.0	57.1	53.1	47.6	42.1	65.4	55.9	45.3	84.5	82.1	80.3	88.8	89.3	88.1
GLM-4.6	53.4	52.6	50.3	49.5	43.9	39.8	58.2	52.0	44.0	84.0	82.3	82.9	85.6	87.2	86.1
Gemini-2.5-Pro	43.9	45.7	42.9	37.7	33.2	30.5	52.1	44.3	38.7	82.5	80.1	78.9	86.2	85.8	84.1
GPT-4.1	42.3	40.8	39.5	35.2	31.8	28.4	48.6	41.2	35.9	63.8	61.5	59.2	84.9	83.7	82.1
Qwen3-235B	44.1	43.5	41.2	38.9	34.7	31.3	50.3	43.8	37.2	52.4	50.8	48.6	85.7	84.9	83.1

D3 Model	IR _{UF}			IR _{SC}			CC			FA			IF		
	64k	128k	512k	64k	128k	512k	64k	128k	512k	64k	128k	512k	64k	128k	512k
Qwen3-32B	38.5	37.2	35.8	32.1	28.5	25.3	44.7	38.2	32.5	58.3	56.1	53.8	80.2	78.9	76.1
Qwen2.5-72B	35.2	34.1	32.5	29.8	26.3	23.1	41.3	35.7	30.2	55.1	53.2	50.8	78.5	76.8	74.1



Extremely Challenging

Best model achieves only 65.6 avg under 512k. Scaling law remains key.



Longer Context = Lower Score

Performance drops as corpus grows 64k→512k. Noise makes evidence harder to find.



IF ≠ FA

Some models achieve good IF but very low FA. Reports look complete but contain factual errors.



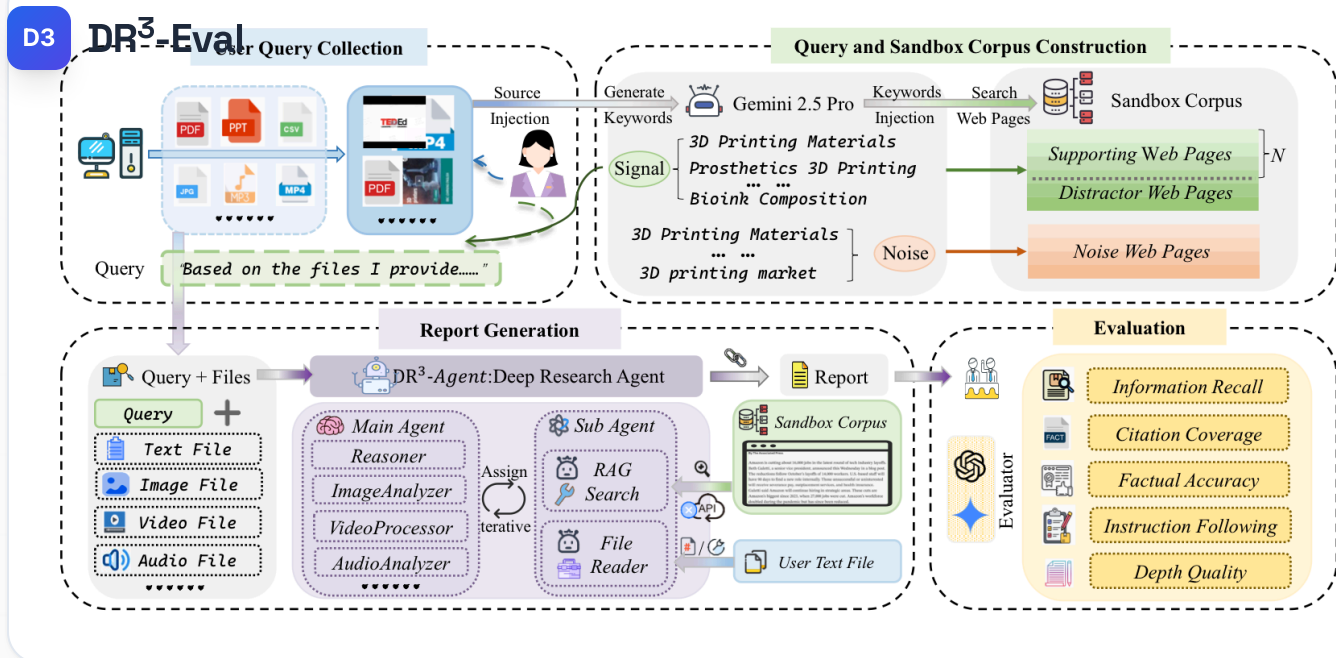
Domain Performance Varies

No single model dominates all 13 sub-domains.



Benchmark Construction

A 5-stage pipeline for realistic, controllable, and precisely evaluable deep research benchmarking.



👤 Stage 1: Real-World Grounding

Volunteers provide multimodal materials (text, images, video, audio). 100 document sets across 3 domains and 13 sub-fields.

🔍 Stage 2: Search Path Distillation

Divergent-convergent keyword generation splits into Signal Keywords (core path) and Noise Keywords (misleading).

🏠 Stage 3: Sandbox Construction

Static sandbox with Supportive, Distractor, and Noise documents. Five context-length settings (32k–512k).

❓ Stage 4: Query Construction

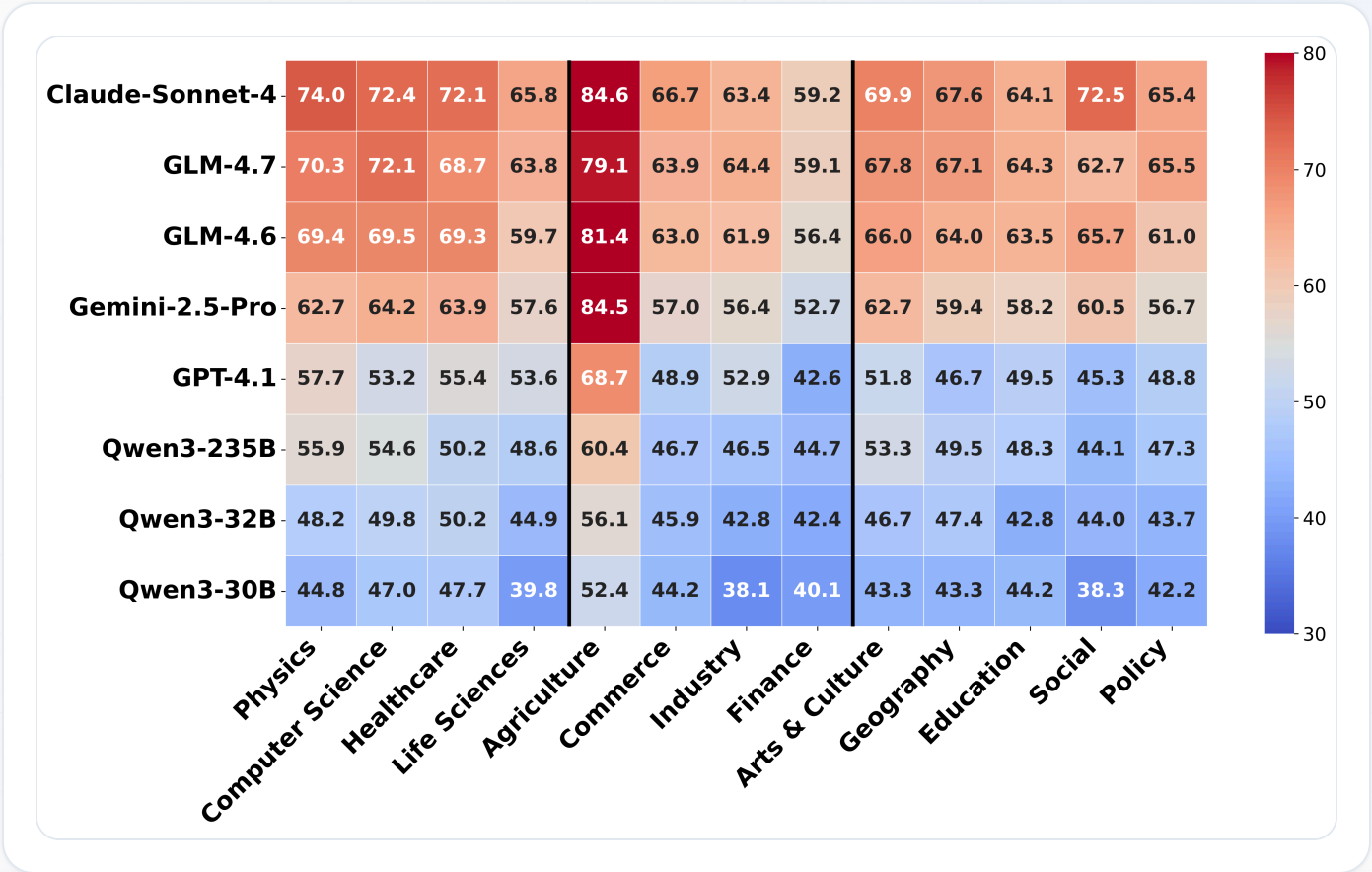
Queries reverse-engineered from verified evidence, ensuring definitive answers grounded in the sandbox.

✅ Stage 5: Quality Control

Four-dimensional validation: implicit guidance, synthesis necessity, insight novelty, interpretative unambiguity.

Result Analysis

Detailed analysis across dimensions, corpus scales, error types, and experimental settings.



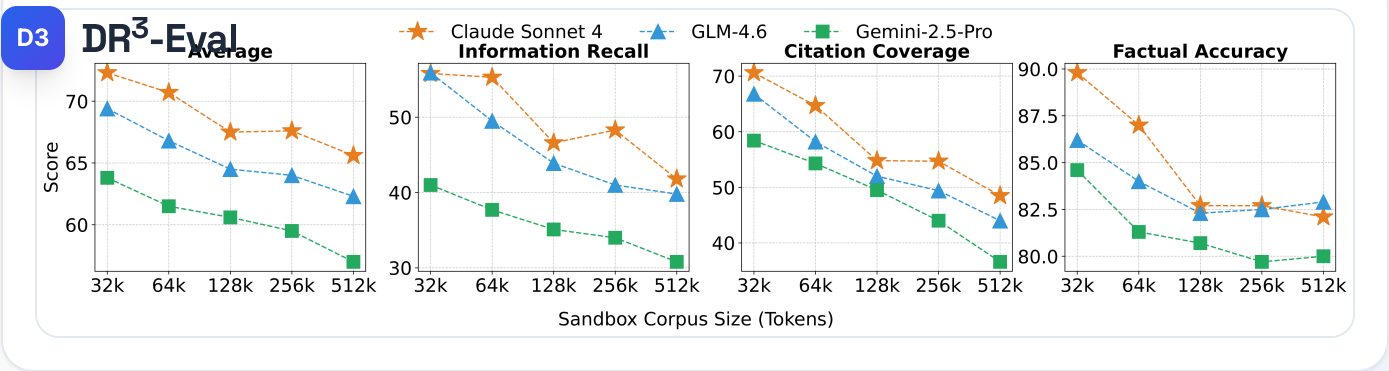
Cross-Domain Heatmap

- Top Performers

Claude Sonnet 4 leads Physics (84.6). GLM-4.7 excels in Industry and Policy.
- Challenging Domains

Agriculture and Commerce are hardest across all models.
- No Universal Winner

No single model dominates all 13 sub-domains.

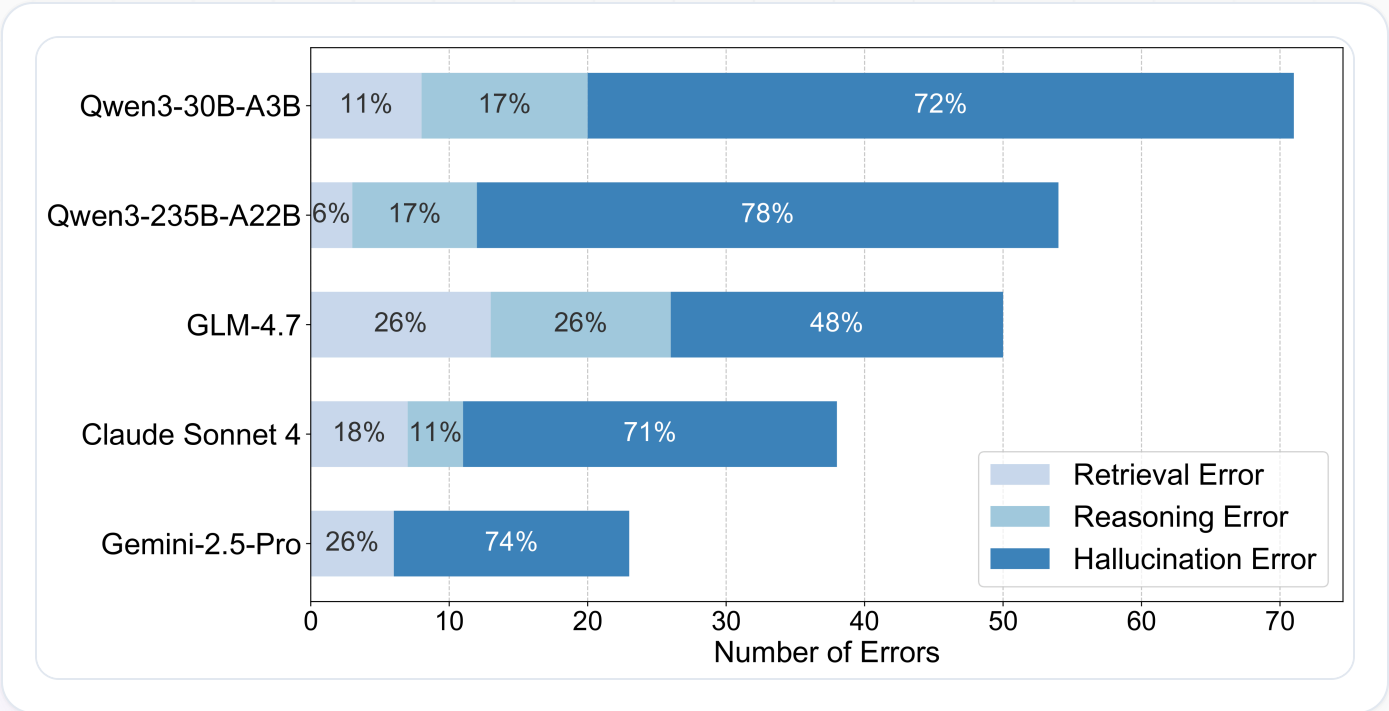


Scale Analysis

Performance Drops
Avg, IR_SC and CC decline as corpus grows from 32k to 512k.

FA Stays Stable
FA remains stable across scales, reflecting reasoning over retrieval.

IF Resilient
IF stays high even at 512k, independent of retrieval difficulty.



Error Type Analysis

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Hallucination Dominates

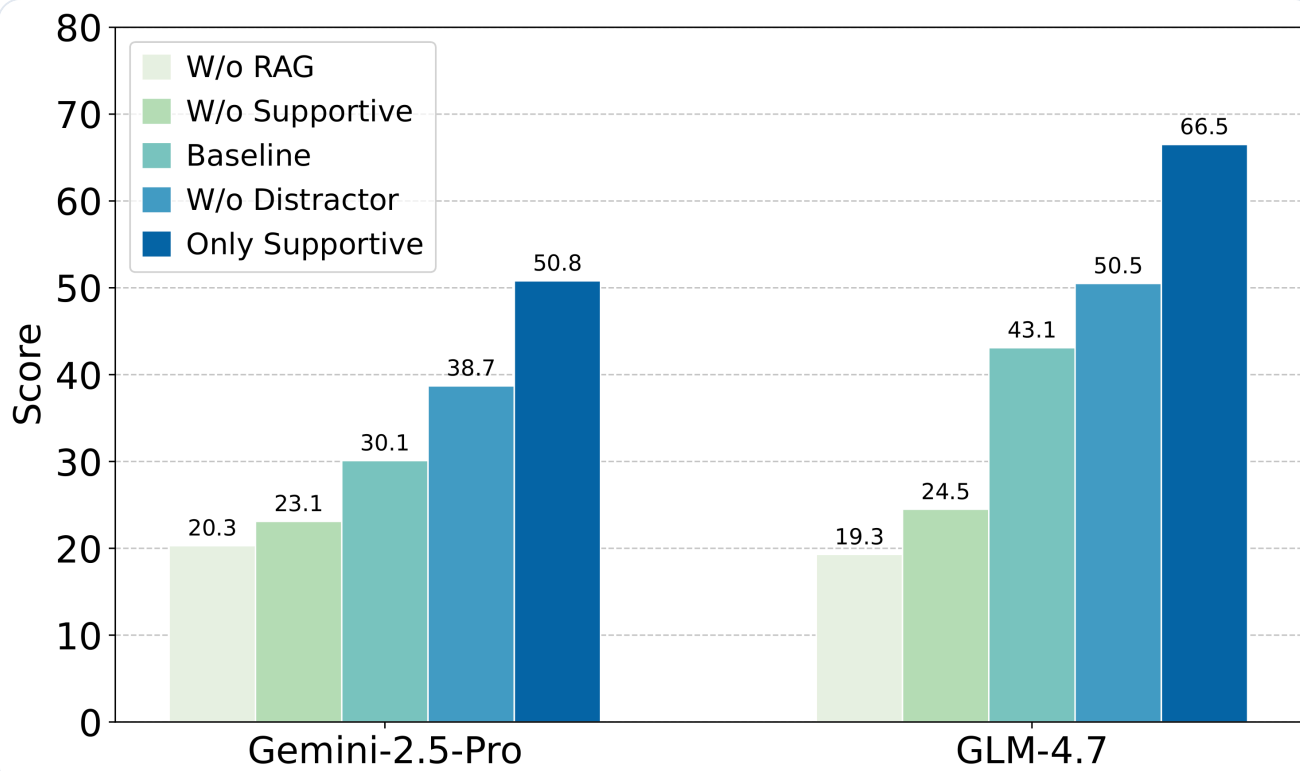
48–78% of errors are hallucinations.

✓ Gemini-2.5-Pro

Lowest hallucination, most retrieval failures.

⚠ Qwen3-235B

78% hallucination despite low retrieval failure.



🔧 Sandbox Corpus Effectiveness

👉 Distractors Work

Removing distractors improves performance significantly.

🔒 No Shortcuts

W/o Supportive $\approx$ W/o RAG, no exploitable shortcuts.

📊 Only Supportive Best

Only supportive docs yield highest scores.

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Metric	Qwen3-235B-A22B			Gemini-2.5-Pro		
	Baseline	w/ Web	Δ	Baseline	w/ Web	Δ
IR _{SC}	31.0	28.1	-2.8	32.8	34.9	+2.1
IR _{UF}	43.6	47.9	+4.3	36.9	42.4	+5.4
FA	63.3	65.9	+2.6	75.6	70.2	-5.4
DQ	67.0	65.5	-1.5	69.5	72.0	+2.5
IF	73.4	79.1	+5.7	81.4	80.4	-1.0
Avg.	54.9	56.6	+1.6	63.1	62.1	-1.0

Sandbox vs. Online

≈ Minimal Difference

Δ < 2 points between sandbox and web search.

Reproducibility

Sandbox simulates open-web with reproducibility.

Model	OpenAI-Emb	Qwen-Emb	BM25
GLM-4.7	56.58	53.61	50.71
GPT-4.1	36.15	35.64	22.60
Gemini-2.5-Pro	49.51	37.16	31.25

Retriever Comparison

Dense Retrieval Wins

OpenAI embedding-3-small achieves best performance.

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 BM25 Falls Behind

Lexical BM25 significantly worse than dense retrieval.

Evaluation Method	Pearson <i>r</i>	Spearman <i>ρ</i>	Pairwise Agr.
DR ³ -Eval (Ours)	0.78	0.73	0.89
Inter-Human	0.83	0.76	0.91

 Human Evaluation

 Strong Alignment

$r=0.78$, $\rho=0.73$, agreement 89%.

 Near Human Level

Approaches inter-human consistency ($r=0.83$, $\rho=0.76$).



Citation

```
@article{dr3eval2025,  
  title={DR3-Eval: Towards Realistic and Reproducible Deep Research Evaluation},  
  author={NJU-LINK Team},  
  journal={arXiv preprint arXiv:xxxx.xxxxx},  
  year={2025}  
}
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